```
#include <stdio.h>
   void main()
   {
      int x = 5;
      if (x < 1)
         printf("hello");
      if (x == 5)
         printf("hi");
      else
         printf("no");
   }
```

What will be the output of the following C code?

ans- hi

```
#include <stdio.h>
   void main()
   {
      int x = 5;
      if (true);
         printf("hello");
   }
```

What will be the output of the following C code?

ans- It will throw an error

```c
#include <stdio.h>

int main() {
    int i;

    for (i = 0; i < 5; i++) {
        if (i % 2 == 0)
            printf("Even\n");
        else if (i % 2 == 1)
            printf("Odd\n");
        else
            printf("Error\n");
    }

    return 0;
}
```

How many times 'Error' will be printed ?

ans- 0 times

```c
#include <stdio.h>

int main() {
    int i;
    int sum = 0;

    for (i = 1; i <= 5; i++);
    {
        sum += i;
    }

    printf("Sum = %d\n", sum);

    return 0;
}
```

What will be the value of ‘sum’ after the loop ?

Sum = 6

```c
#include <stdio.h>

int main() {
    int i;
    int sum = 0;

    for (i = 1; i <= 5; i++) {
        switch (i) {
            case 1:
            case 3:
            case 5:
                sum += i;
                break;
            case 2:
                sum -= i;
                break;
            case 4:
                sum *= i;
        }
    }
    printf("Sum = %d\n", sum);

    return 0;
}
```

what is the value of 'sum' after the loop ?

Ans- Sum = 13

```c
#include <stdio.h>

int main() {
    int i, result = 0;

    for (i = 1; i <= 4; i++) {
        if (i % 2 == 0) {
            result += i;
        } else if (i % 3 == 0) {
            result *= i;
        } else {
            result -= i;
        }
    }

    printf("Result = %d\n", result);

    return 0;
}
```

Result = ?

Result = 7

```c
#include <stdio.h>

int main() {
    int i, j, sum = 0;

    for (i = 1, j = 5; i <= 5 && j > 0; i++, j--) {
        if (i == j)
            sum += i * j;
        else if (i < j)
            sum += i;
        else
            sum += j;
    }

    printf("Sum = %d\n", sum);

    return 0;
}
```

Sum = ?

Sum = 15

```c
#include <stdio.h>

int main() {
    int x = 2, y = 1;
    while (x < 4) {
        if (x % 2 == 0) {
            switch (y) {
                case 1:
                    y++;
                    break;
                case 2:
                    y--;
                    break;
            }
        } else {
            y += x;
        }
        x++;
    }

    printf("x = %d, y = %d\n", x, y);

    return 0;
}
```

x = ?, y = ?

x = 4, y = 5

```c
#include <stdio.h>

int main() {
    int x;

    if (x > 0)
        printf("Positive\n");
    else
        printf("Non-positive\n");

    return 0;
}
```

What will be the output ?

Answer: Undefined Behavior (Explanation: The variable x is used without being initialized, leading to undefined behavior.)

```c
#include <stdio.h>

int main() {
    int i = 0;

    for (int i = 0; i < 3; i++) {
        printf("%d ", i);
    }

    printf("%d\n", i);

    return 0;
}
```

What will be the output?

0 1 2 0 (Explanation: The i inside the for loop is a separate variable from the i declared outside the loop. The loop prints 0 1 2, and the outer i remains 0.)

```c
#include <stdio.h>

int main() {
    int x = 5;

    int y = x++ + ++x;

    printf("y = %d\n", y);

    return 0;
}
```

What will be the value of y?

y = 12 (Explanation: The expression x++ uses the value of x before incrementing, while ++x increments x first. So, y = 5 + 7 = 12.)

Quiz

1. Write a program to input 4 numbers and print out the largest and the second largest number.

2. Write a program to input a 3 digit number and reverse it. And check if the number is palindrome or not.

3. Write a C program to input any character and check whether it is a number or a alphabet or neither any of those.